

TITLE: The Prevention and Treatment of ICU Delirium

PROTOCOL: ICU Delirium

PURPOSE: The purpose of the ICU Delirium protocol is to reduce the incidence, duration and adverse outcomes associated with delirium in critically ill patients. Delirium in the ICU is often multifactorial, and interventions are necessary to improve patient outcomes and mitigate adverse effects of delirium.

APPROVED:

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DEFINITIONS:

- A. **Acute Encephalopathy** (also referred to as “Fluctuating Encephalopathy”) – rapidly developing (<4 weeks) mental status change resulting from a pathobiological process in the brain. Clinically includes Acute Coma, Delirium, and Subsyndromal Delirium, as well as other diagnoses not discussed in this protocol.
- B. **Delirium** – a clinical state with all the following features, as described by DSM-5 criteria
 - a. Both of the following:
 - i. Disturbance in Attention – inability to direct, focus, sustain, and shift attention
 - ii. Disturbance in Awareness – reduced orientation to the environment
 - b. The disturbance develops over a short period of time (usually hours to days), represents an acute change from baseline, and tends to fluctuate in severity over the course of the day
 - c. An additional disturbance in cognition is present
 - i. Memory
 - ii. Orientation
 - iii. Language
 - iv. Perception
 - v. Visuospatial ability
 - d. The disturbances are not better explained by a pre-existing, established, or evolving alternative neurocognitive disorder, and do not occur in the context of a severely reduced level of arousal such as coma
 - e. There is evidence from either the history, physical examination, or laboratory findings that the disturbance is a direct physiologic consequence of another medical condition known to contribute to delirium, or multiple etiologies
- C. **Subsyndromal Delirium (SSD)** – An acute change from baseline mental status that does not meet criteria for delirium, but includes one or more core symptoms/criteria for delirium, and may or may not ultimately progress to a diagnosis of delirium
- D. **At risk for delirium** – patients who are more likely to develop either SSD or delirium. (see Appendix A)
- E. **Coma** – a state characterized by the absence of both arousal and awareness of one’s environment. Usually, prerequisite is a RASS score of either -4 or -5
- F. **Excited delirium** – a medical emergency characterized by profound encephalopathy, agitation, hyperthermia and sympathetic nervous system activation, potentially requiring induction of anesthesia and pharmacologic paralysis for treatment. This is a rare diagnosis outside of the emergency department and medical ICU and is not covered in the below tiered approach to treatment.

CORE PROCEDURE:

NURSING RESPONSIBILITIES

- General
 - Identify at-risk patients (see Appendix A)
 - All patients must have a daily spontaneous awakening trial or sedation holiday, unless contraindicated
 - Examples of contraindications:
 - Profound hemodynamic instability requiring several vasoactive infusions
 - Traumatic Brain Injury with active management of elevated ICP
 - ARDS or profound hypoxemia requiring careful mechanical ventilation management
 - Status epilepticus actively undergoing induction of ASM therapy
 - Any disease condition requiring pharmacologic paralysis
 - Early mobilization unless contraindicated
 - Encourage Family Visitation
 - Adherence to all tier 0 therapies (described below) unless not possible
- Patient Assessment
 - Screen for delirium using the Confusion Assessment Method for the ICU (CAM-ICU) every nursing shift (twice daily, assess during spontaneous awakening trial, if applicable)
 - Document screening results in the electronic medical record
 - Report delirium positive results to the physician, nurse practitioner or physician's assistant during rounds
 - Assess for other causes of neurological status changes (including but not limited to: hypoxia, stroke, seizure, infection, hypoglycemia, over/undertreated pain, etc.)
 - Assess patients for extrapyramidal symptoms (including acute dystonia, akathisia, parkinsonism, tardive dyskinesia), drug-induced rigidity, and fever at least twice daily (every nursing shift)

TIERED APPROACH TO ICU DELIRIUM AND ASSOCIATED SYNDROMES

What follows is a tiered approach to the prevention and management of delirium and associated syndromes. A few important points follow:

- **The most effective treatment of delirium is time!** The vast majority of the pharmacologic agents below have not been demonstrated to improve duration of delirium symptoms, ICU length of stay, mortality or other patient-centered outcomes.
- Each tier is separated into pharmacologic interventions and nonpharmacologic interventions.
 - **Nonpharmacologic interventions** are recommended for all patients in their respective tiers.
 - **Pharmacologic interventions** simply provide suggestions for therapies to consider for these patients, recognizing that a one-size-fits-all approach has never been reported in literature and may not be the best way to manage delirium. Provider discretion is often a large driver of agent selection
 - It is generally recommended to max out the dosing of a single agent before adding an additional agent
- The tiered approach requires that **early tier interventions are addressed and/or initiated before moving on to the next tier.**
- **Tiers 3 and 4 utilize more aggressive therapies** intended for patients with hyperactive features, and are more likely to contribute to iatrogenesis. These therapies must be de-escalated once delirium resolves or if hypoactive features develop
- This tiered approach is not intended for use in patients with excited delirium, which is a medical emergency and may require aggressive use of anesthetics and/or pharmacologic paralysis
- All patients with pharmacologic interventions should have their QTc monitored on telemetry, and/or with occasional EKG based on telemetry findings and/or individual provider discretion. The goal is to recognize patients at risk for Torsades De Pointes (TdP). Antipsychotic induced TdP is rare. **Patients with QTc < 500 are highly unlikely to develop TdP.**
 - In patients with a prolonged QTc > 500, use of medications at high risk for inducing TdP should be reconsidered, especially in patients at high risk of TdP
 - Patient conditions with high risk of developing TdP
 - Hypomagnesemia, Hypokalemia
 - History of LQTS or baseline long QT interval

- Bradycardia
- Recent cardioversion
- Concomitant use of antidysrhythmic agents which are known to be associated with TdP
- Medications high risk for inducing TdP include haloperidol, ziprasidone, quetiapine. **IV medications are of highest risk**
- Medications considered low risk for TdP include olanzapine, aripiprazole, trazodone.

INTERVENTIONS

1. **Tier 0** – these interventions are intended for all patients in the ICU, regardless of risk of delirium or diagnoses. Interventions in this tier differ during “daytime” versus “nighttime” hours, and are designed to prevent delirium from developing, and to promote quality ICU sleep hygiene and daytime activity
 - a. **Daytime Hours 6 am – 10 pm**
 - i. Encourage wakefulness behavior, discourage “naps”
 - ii. Offer and provide adequate daytime stimulating activities
 - iii. Early mobilization unless contraindicated
 - iv. Keep overhead and bedside lights on, and open blinds
 - v. Encourage family involvement in wakefulness activity, and instruct family on delirium prevention measures
 - vi. Frequent re-orientation of patient to date, time, place
 - b. **Nighttime Hours 10 pm-6 am**
 - i. Turn off all overhead, bedside and window lights in the room
 - ii. Lower blackout curtains, close shades
 - iii. Turn off the TV
 - iv. Provide sleep mask, earplugs, and other sleep hygiene aids
 - v. Minimize patient disturbances unless medically necessary
 - vi. Consider postponing blood draws until 6am, unless medically necessary to draw earlier
 - vii. Deescalate nighttime neurologic examinations as able, depending on the relevant diagnoses
 - viii. Consider quiet time orders for select patients
 - c. **All patients**
 - i. Daily spontaneous breathing and awakening trials
 - ii. Ensure pain is adequately treated, but not overtreated. Use an evidence-based pain scale to determine pain management needs
 - iii. Minimize sedation of intubated patients to target a desired RASS goal
 - iv. Daily assessment and aggressive management of polypharmacy syndromes
 - v. Restart indicated home psychotropic medications as early as possible
 - vi. Screen for and treat withdrawal syndromes as appropriate

- vii. Deescalate neurologic checks
 - viii. Ensure PT/OT is consulted on all indicated patients
 - ix. Pursue early mobilization on all patients, unless contraindicated
 - x. Remove invasive devices and lines once clinically appropriate
 - xi. Screen for delirium with CAM-ICU, recognizing its limitations, including lower than previously reported sensitivity, especially in patient populations with concomitant neurologic disease¹
 - 1. Sensitivity 80%
 - 2. Specificity 95%
2. **Tier 1** – These interventions are intended to provide an extra layer of both nonpharmacologic and pharmacologic therapy for patients **at high risk** for developing delirium or associated syndromes. Interventions may also benefit **SSD** patients early in the course of the disease. Pharmacologic interventions only include medications intended for use as sleep aids. See appendix A for high-risk patient factors

a. Nonpharmacologic Interventions

i. Address, diagnose, and treat conditions known to contribute to delirium

- 1. Undertreated pain
- 2. Hypo/Hyponatremia
- 3. Hypo/hypercalcemia
- 4. Uremia and AKI
- 5. Infection
- 6. Hypo/hyperglycemia
- 7. Constipation
- 8. Intracranial pathology
- 9. Seizures/ictal-interictal continuum
- 10. Medication side effects and neurotoxicities
- 11. Other Electrolyte Abnormalities
- 12. Hypercapnia
- 13. Acid-base disturbances
- 14. Vitamin deficiencies

ii. Minimize or discontinue medications known to contribute to delirium

- 1. Benzodiazepines** (lorazepam, alprazolam, clonazepam)
- 2. Anticholinergics/Antihistamines** (hydroxyzine, diphenhydramine)
- 3. Beta-lactams/FQs/Bactrim
- 4. Cefepime
- 5. H2 blockers

6. Antiseizure medications
7. Antispasmodics
8. Gabapentin/Pregabalin
9. Antidopaminergics
10. Zolpidem
11. Steroids
12. Polypharmacy
13. If AKI develops, be sure that medications are re-dosed based on creatinine clearance

** highly delirium-inducing, and should be avoided in most cases

iii. Consider geriatric consultation for elderly patients

iv. Consider early Palliative care consult for patients with disease-related anxiety or psychiatric distress

b. Pharmacologic Interventions

i. Restart Home Psychiatric Medications

- ii. Consider starting Sleep Aid** - Be sure sleep medication orders are such that they are given at the proper time. When ordered as Prn, sleep medications may be unintentionally given during early AM hours, and may lead to perpetuation of sleep/wake cycle disturbances

1. Melatonin 6-9mg qHS
2. Trazodone 25-200mg qHS
3. Mirtazapine 7.5mg qHS

3. Tier 2 – Suspected Delirium – this tier is intended for patients who have been diagnosed with syndromal delirium based on DSM-V criteria (see above definition), or patients who screen as CAM-ICU positive. This tier can also be considered for patients with subsyndromal delirium refractory to tier 1 therapies.

a. Consider workup to rule out alternative diagnoses

- i. Delirium is highly prevalent, and far more common in an ICU setting than alternative etiologies, and a workup may not always be necessary.
 1. The neurologic exam is an important guide to selecting appropriate diagnostics
- ii. See Appendix B for possible diagnostic studies to consider in select cases

b. Optimize Tier 0 and 1 therapies

c. Pharmacologic Interventions

i. RASS 0-5

1. Dexmedetomidine infusion 0.1-0.5 mcg/kg/min initially
 - a. Strongest evidentiary support compared to other agents
 - b. RCT level data never exceeded dosing of 0.7mcg/kg/min
 - c. Can be escalated to max dose of 1.5mcg/kg/min if intubated
 - d. Higher doses increase the risk of microaspiration and oversedation
2. Olanzapine 2.5-5 mg PO or ODT
 - a. IM acceptable unable to take ODT or PO route
3. Quetiapine 25-100mg PO
4. Mirtazapine 7.5-30mg qHS

ii. RASS <0

1. **See Delirium with Hypoactive Features section below**

- 4. Tier 3 –** Interventions in this tier are intended for patients with delirium refractory to Tier 0-2 interventions and are mostly intended for patients with agitation. Escalation of pharmacology is the mainstay of treatment. These medications should be given via an enteral or sublingual route of administration whenever possible

a. Optimize Tier 0-2 therapies

- b. Pharmacologic Interventions –** must balance risk of oversedation with risk of delirium symptoms causing direct harm to the patient

i. RASS 1-5

1. Precedex infusion 0-0.7mcg/kg/min
 - a. Escalate to higher than these doses in non-intubated patients with extreme caution
2. Olanzapine (HS or BID) – daily max 20mg
3. Quetiapine (HS, BID or TID) – daily max 600mg
4. Aripiprazole 5-30mg HS
5. Valproic acid 125-250mg BID or TID (20mg/kg/day max)
6. Guanfacine 0.5mg-2mg HS or BID

ii. RASS <0

1. **See Section on Delirium with Hypoactive Features Below**

c. Consider psychiatry consult for comanagement

5. **Tier 4** – It is rare that a patient should require medications in this tier. These interventions are intended for use in patients with worsening delirium syndromes despite maximization of tier 3 therapy. Patients with severe agitation that pose an immediate threat to self or staff may also benefit from this tier of interventions
 - a. **Optimize Tier 0-3 therapies**
 - b. **Pharmacologic Interventions** – must balance risk of oversedation with risk of delirium symptoms causing direct harm to the patient
 - i. **RASS > 2**
 1. **Rescue medications** – typically should not have standing orders
 - a. Olanzapine 2.5mg- 10mg/dose IM
 - b. Haloperidol 2mg-5mg/dose IV or IM
 - c. Droperidol 0.625-2.5mg/dose IV
 - d. Ziprasidone 10-20mg/dose IM
 2. **Maintenance medications**
 - a. Haloperidol 2mg-5mg/dose PO
 - b. Pregabalin 75-150mg TID
 - ii. **RASS 1-2** – strongly reconsider need for tier 4 therapies
 - iii. **RASS <1** – tier 4 therapies not indicated
 - c. **Call Security for help if patients are a threat to staff, or require initiation of physical restraints**
 - i. Remove locked restraints as soon as is safe
 - d. **Consider psychiatry consult for comanagement**
6. **Daily Assessment for Polypharmacy** – once delirium has been resolved, and/or agitation has improved to a safe level, de-escalation of tier 3 and tier 4 therapies should commence.
 - a. Daytime sleepiness should prompt immediate de-escalation, as this can compromise mobility, sleep/wake cycle regulation, and paradoxically worsen delirium. Furthermore, most tier 2-4 medications are considered potentially delirium-inducing.
7. **Delirium with Hypoactive Features** – Defined as delirium with a RASS < 0, hypoactive delirium should be treated differently than hyperactive delirium throughout tier 2-tier 4. The following is how treatment should differ:

- i. Consider extended workup for alternative etiologies of encephalopathy
 - 1. See Appendix B
- ii. Continue Tier 0 and Tier 1 approach for all delirious patients with hypoactive features, including consideration of addition of a sleep aid
- iii. Discontinue daytime sedating medications, including tier 2-4 therapy administered during wakeful hours
- iv. Consider a pharmacological minimalist approach – there is currently no quality literature to suggest that addition of pharmacology improves these patients
- v. If hypoactive delirium is refractory, consider a neurostimulant trial - usually lasting at least 1-2 weeks
 - 1. Modafinil 100-200mg daily
 - a. Targets arousal pathways
 - b. Safe, swiss-army knife like drug
 - c. Minimal side effects
 - 2. Amantadine 50-200mg BID (6am and noon)
 - a. More effective with structural disease (i.e. frontal TBI)
 - b. Works quickly
 - c. Lowers seizure threshold – avoid in seizure predisposition or IIC
 - d. Nausea, vomiting
 - 3. Methylphenidate – 2.5mg BID
 - a. Recommend PMR or Neurology consult if considering
 - b. Targets arousal pathways
 - c. Causes sympathetic outflow
 - d. Rapid onset
 - 4. Sinemet – 25/100mg tabs – 1 tablet BID
 - a. Recommend PMR or Neurology consult if considering
 - b. Drug induced parkinsonism
 - c. Abulia, akinetic mutism

APPENDIX A – risk factors for delirium

Baseline Risk Factors

- History of Dementia
- Advanced age (2% increased incidence per year after age 64)
- Depression and anxiety
- Smoking history
- Alcohol or substance abuse history
- Acute or Chronic pain
- Higher severity of illness on hospital admission

Acquired risk factors

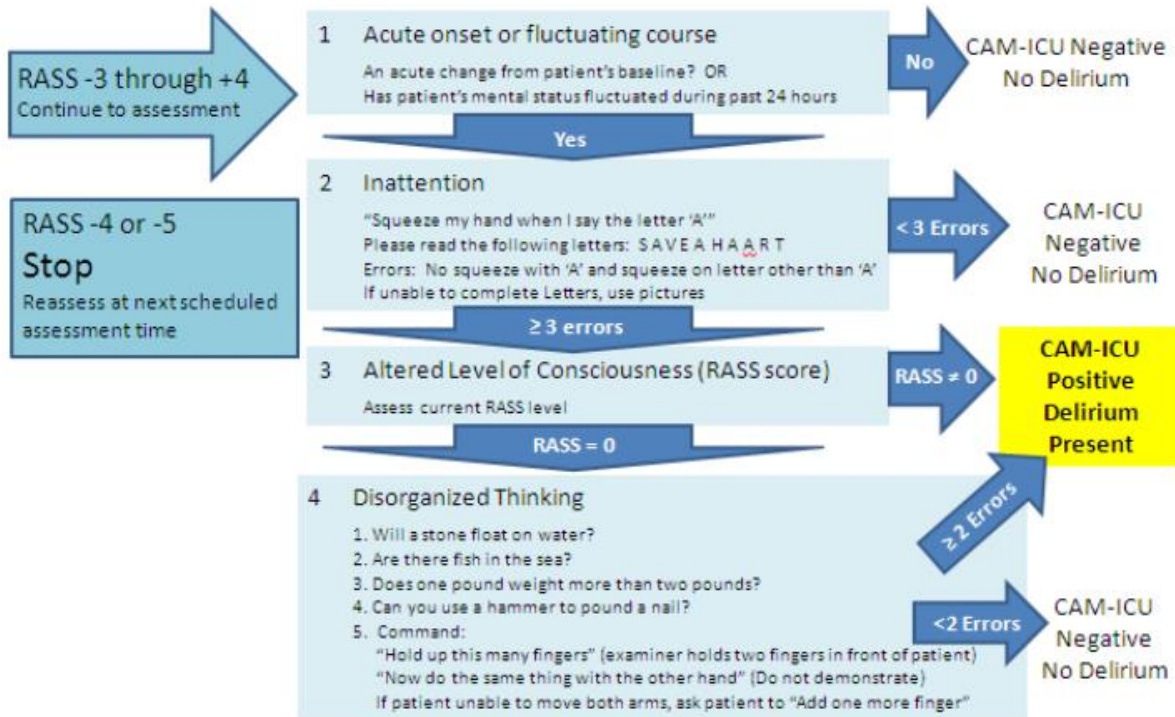
- Hypoxia
- Metabolic disturbances
- Electrolyte abnormalities
- Infection
- Prolonged restraint use and immobility
- Withdrawal syndromes
- Seizures/IIC
- Dehydration
- Hypo/hyperthermia
- Head trauma
- Post neurosurgical procedure
- Vascular disorders
- Medication effects
- Vitamin Deficiencies

APPENDIX B – Considerations for the workup of acute encephalopathy

- Comprehensive Neurologic exam
 - Orientation questions
 - GCS (low yield) and/or Four Score
 - Sensorimotor exam
 - Cranial nerve exam
 - Deep tendon reflexes, and assess for clonus
 - Passive range of motion to assess for rigidity
 - Assess for involuntary movements
- Neuroimaging
 - Noncontrast CT brain is reasonable in most patients
 - CTA brain if there is concern for a focal deficit on exam
 - MRI brain can be considered in the right clinical context for refractory cases
- EEG
 - >12 hours recommended for most patients
 - >24 hours with event suspicious for seizure
- Extended laboratory panel
 - Ionized calcium
 - TSH with Free T4
 - Ammonia
 - VBG/ABG
 - UA
- Drug monitoring levels
 - Immunosuppressants
 - Antiseizure Medications
 - Theophylline
 - Salicylate/acetaminophen
- CSF studies
 - ICU-acquired meningitis is very unusual, but can be considered in select post-surgical or immunocompromised patients, or patients with endocarditis
- Urine toxicology screen is expensive, is not capable of providing a definitive diagnosis, and is highly unlikely to be useful

Appendix C – CAM ICU¹

Confusion Assessment Method for the ICU



Adapted from: E. Wesley Ely, MD, MPH and Vanderbilt University, Copyright © 2002, all rights reserved
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- Recent literature suggests that CAM-ICU may be less effective than previously thought, especially in patients with either concomitant structural or organic neurologic disease. Thus, in patients with a high pretest probability for delirium, a negative CAM-ICU screen may lack adequate negative predictive value to be an effective rule out tool.¹

Appendix D – RASS scale²

Score	Term	Description
+4	Combative	Overtly combative or violent; immediate danger to staff
+3	Very agitation	Pulls on or removes tube(s) or catheter(s) or has aggressive behavior toward staff
+2	Agitated	Frequent nonpurposeful movement or patient-ventilator dyssynchrony
+1	Restless	Anxious or apprehensive but movements not aggressive or vigorous
0	Alert and calm	
-1	Drowsy	Not fully alert, but has sustained (more than 10 seconds) awakening, with eye contact, to voice
-2	Light sedation	Briefly (less than 10 seconds) awakens with eye contact to voice
-3	Moderate sedation	Any movement (but no eye contact) to voice
-4	Deep sedation	No response to voice, but any movement to physical stimulation
-5	Unarousable	No response to voice or physical stimulation

Procedure

1. Observe patient. Is patient alert and calm (score 0)?
Does patient have behavior that is consistent with restlessness or agitation (score +1 to +4 using the criteria listed above, under DESCRIPTION)?
2. If patient is not alert, in a loud speaking voice state patient's name and direct patient to open eyes and look at speaker. Repeat once if necessary. Can prompt patient to continue looking at speaker.
Patient has eye opening and eye contact, which is sustained for more than 10 seconds (score -1).
Patient has eye opening and eye contact, but this is not sustained for 10 seconds (score -2).
Patient has any movement in response to voice, excluding eye contact (score -3).
3. If patient does not respond to voice, physically stimulate patient by shaking shoulder and then rubbing sternum if there is no response to shaking shoulder.
Patient has any movement to physical stimulation (score -4).
Patient has no response to voice or physical stimulation (score -5).

KEYWORDS:

Delirium, subsyndrome delirium, acute encephalopathy, antipsychotic, hyperactive, hypoactive

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